Rubber_band_analysis

Analysis of Field Data and AI-AI in Rubber-band dynamics

PENPAL Project

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0.1 Summary

- Across both datasets, early–middle valence gaps strongly predict opposite-direction changes from middle–late (negative slope).
- This reveals a robust "rubber-band" effect: emotional divergence later snaps back toward each participant's initial baseline.
- The pattern holds for short and long interaction sequences; interaction volume does not moderate the effect.
- The rebound is slightly stronger in the field data, but present in both human–AI and AI–AI conditions.
- This suggests an intrinsic stabilising tendency in the model that humans weakly amplify rather than disrupt.

1 1. Data Loading and Preprocessing

1.1 1.1 Load Data

```
## Field data dimensions: 820 14
```

```
## Simulated data dimensions: 30 12
```

```
##
## Field data columns:
```

```
##  [1] "timestamp"            "user"                 "ai"
##  [4] "conversation_id"      "respondent_id"        "interaction_count"
##  [7] "llm_type"             "turn"                 "starter"
## [10] "user_embedding"       "ai_embedding"         "pct_turn"
## [13] "user_sentiment_score" "ai_sentiment_score"
```

```
##
## Simulated data columns:
```

```
##  [1] "turn"                 "user"                 "ai"
##  [4] "client_id"            "workshop_id"          "conversation_id"
##  [7] "timestamp"            "user_embedding"       "ai_embedding"
## [10] "pct_turn"             "user_sentiment_score" "ai_sentiment_score"
```

1.2 1.2 Data Preprocessing

```
##
## === Field Data Summary ===
```

```
## # A tibble: 1 x 5
##   n_conversations n_turns mean_turns_per_story mean_user_valence mean_ai_valence
##             <int>   <int>                <dbl>             <dbl>           <dbl>
## 1              81     729                    9            0.0426           0.171
```

```
##
## === Simulated Data Summary ===
```

```
## # A tibble: 1 x 5
##   n_conversations n_turns mean_turns_per_story mean_user_valence mean_ai_valence
##             <int>   <int>                <dbl>             <dbl>           <dbl>
## 1               6      30                    5            -0.407          -0.296
```

1.3 1.3 Compute Lagged Variables and Cosine

```
## === Lag Variable Verification ===
```

```
## Field data - Turn 1 with non-NA lag (should be 0): 0
```

```
## Simulated data - Turn 1 with non-NA lag (should be 0): 0
```

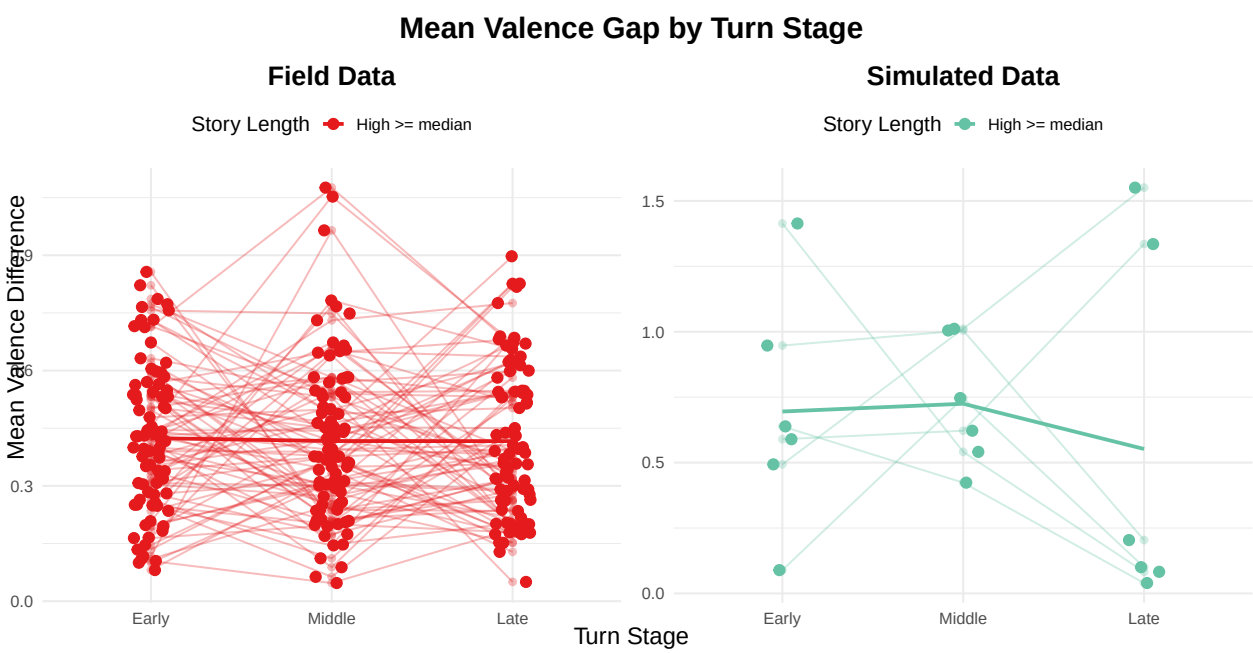
2 2. Temporal Dynamics Analysis

2.1 2.1 Stage-Based Analysis (Early/Middle/Late)

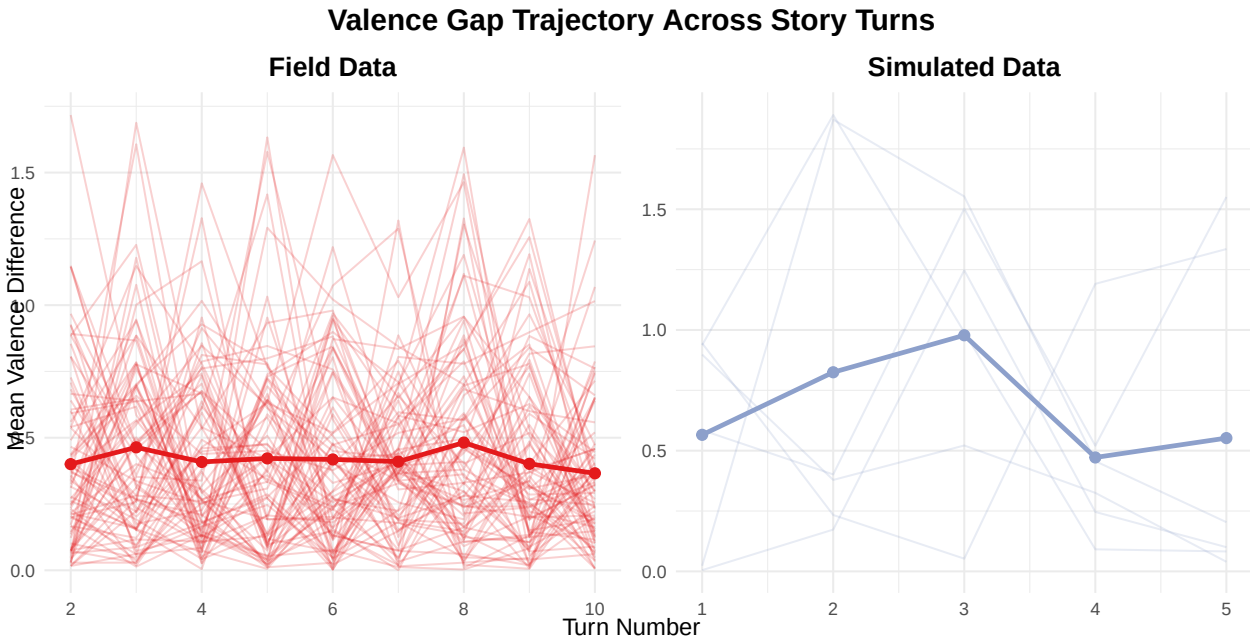
```
## === Mean Valence Gap by Stage ===
```

```
## # A tibble: 6 x 4
##   data_source stage_label mean_gap sd_gap
##   <chr>       <fct>          <dbl>  <dbl>
## 1 Field       Early          0.424     NA
## 2 Field       Middle         0.417     NA
## 3 Field       Late           0.416     NA
## 4 Simulated   Early          0.695     NA
## 5 Simulated   Middle         0.725     NA
## 6 Simulated   Late           0.552     NA
```

2.2 2.2 Stage Valence Gap Plot



2.3 2.3 Turn-by-Turn Valence Gap Trajectory



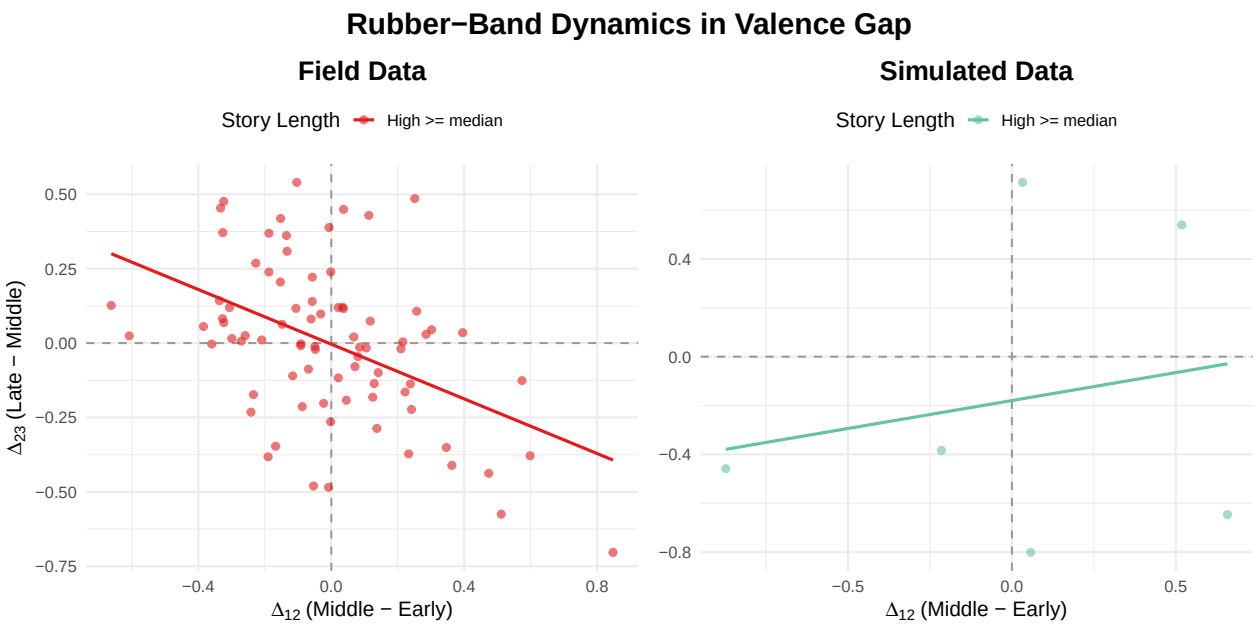
2.4 2.4 Rubber-Band Dynamics

```
## === Rubber-Band Model: Field Data ===

##                 Estimate Std. Error    t value     Pr(>|t|)
## (Intercept) -0.003751029 0.02606538 -0.1439085 8.859392e-01
## delta_12    -0.459356312 0.09879818 -4.6494410 1.313973e-05

##
## === Rubber-Band Model: Simulated Data ===
```

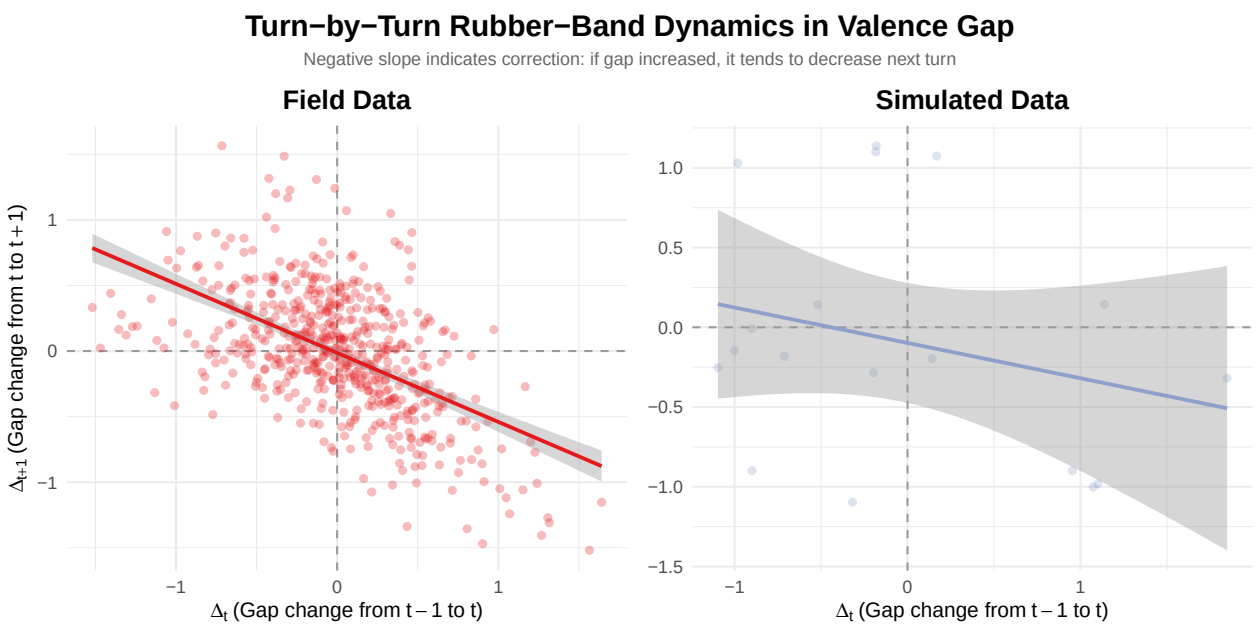
2.5 2.5 Turn-by-Turn Rubber-Band Dynamics

```
## === Turn-Level Rubber-Band Model: Field Data ===

##               Estimate Std. Error     t value     Pr(>|t|)
## (Intercept) -0.0138917 0.01675561  -0.8290775 4.074107e-01
## gap_change  -0.5262275 0.03509412 -14.9947506 4.956090e-43

##
## === Turn-Level Rubber-Band Model: Simulated Data ===

##                Estimate Std. Error    t value  Pr(>|t|)
## (Intercept) -0.09782656  0.1775206 -0.5510717 0.5891977
## gap_change  -0.22171240  0.2027281 -1.0936443 0.2903035

##
## === Turn-Level Rubber-Band Correlation ===

## Field correlation (gap_change vs gap_change_next): -0.5335425

## Simulated correlation (gap_change vs gap_change_next): -0.2637313
```

2.6 2.6 Rubber-Band Dynamics Plot



2.7 2.7 Turn-Level Rubber-Band Dynamics Plot



3 3. Permutation tests

3.1 3.1 Stage-level Permutation Test

```
## === Stage-Level Permutation Test Results ===
## Observed slope: -0.4594
## Null mean: -0.5001
## Null SD: 0.0988
```

```
## Effect size (z): 0.41
## P-value: 0.649
```

3.2 3.2 Brutal Stage-Level Permutation

```
## === Stage-Level Permutation Test Results - Shuffle Turn-Level Valence Gap first ===
## Observed slope: -0.4594
## Null mean: -0.5022
## Null SD: 0.1024
## Effect size (z): 0.42
## P-value: 0.67
```

3.3 3.2 Global Shuffle Stage-Level Test

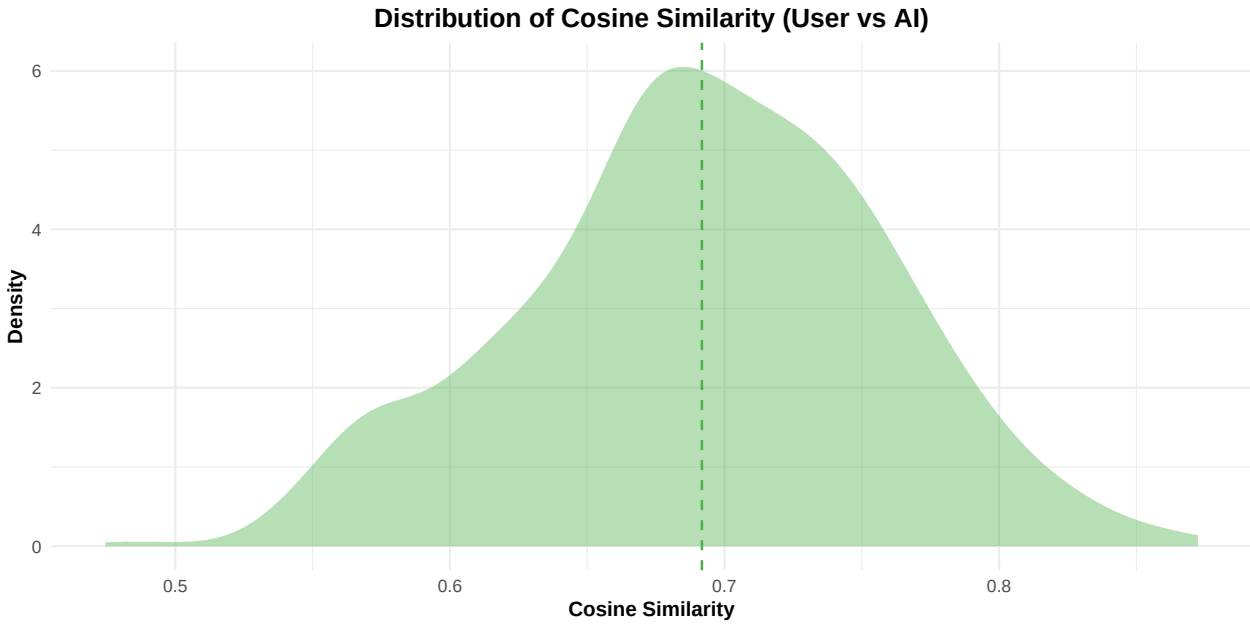
```
##               Estimate Std. Error   t value   Pr(>|t|)
## (Intercept)  0.0662683 0.05871874  1.128572 0.27312777
## d12         -0.6487350 0.33162040 -1.956258 0.06529781
```

3.4 3.3 Turn-Level Permutation Test

```
## === Turn-Level Permutation Test Results ===
## Observed slope: -0.5262
## Null mean: -0.0025
## Null SD: 0.0577
## Effect size (z): -9.08
## P-value: <2e-16
## # A tibble: 2 x 4
##   transition_type rb_slope correlation     n
##   <chr>              <dbl>       <dbl> <int>
## 1 Stage Boundary    -0.516      -0.501   162
## 2 Within-Stage      -0.564      -0.579   324
```

4 4. Cosine similarity alignment

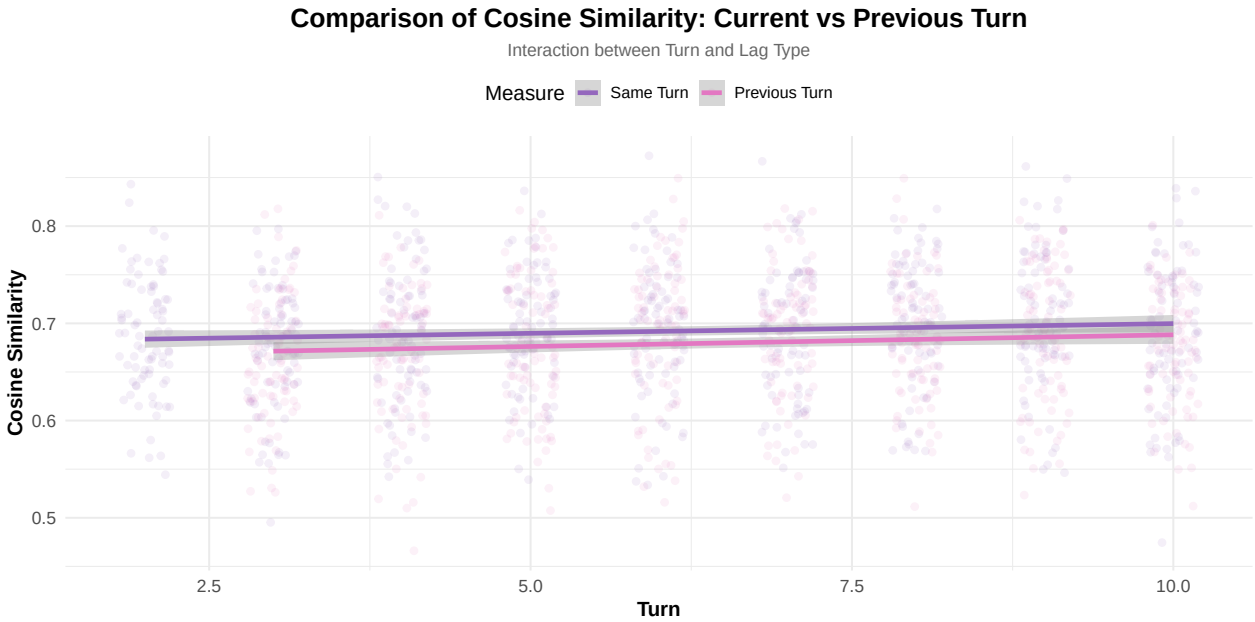
4.1 4.1 Cosine Similarity Distribution



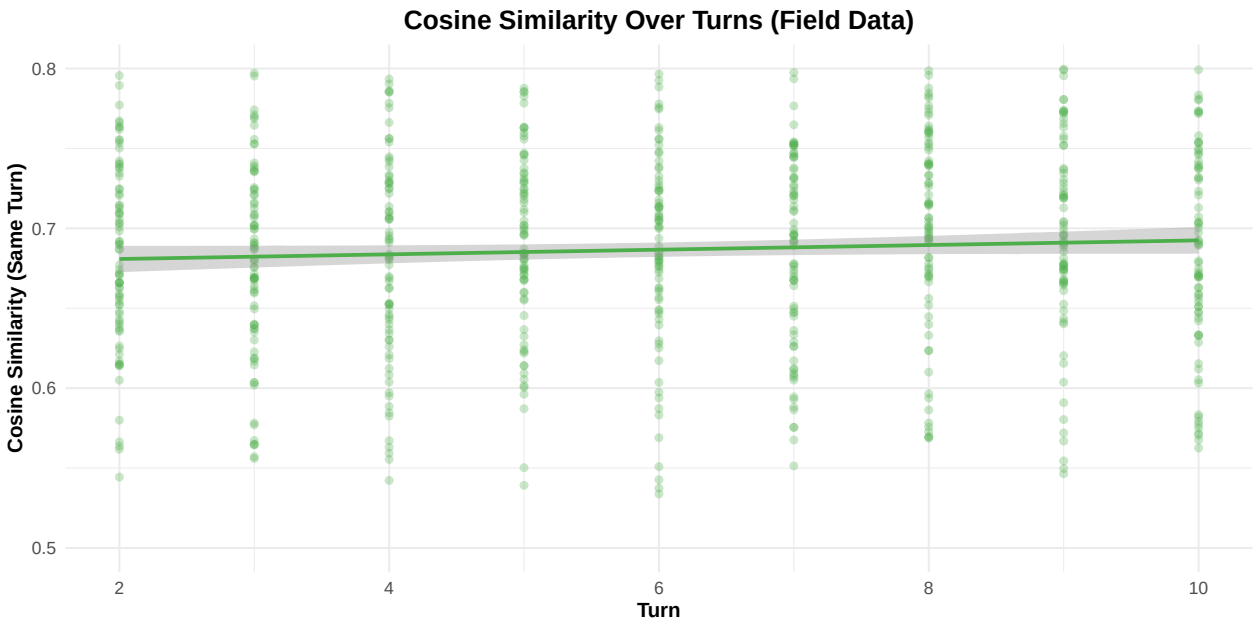
4.2 4.2 Cosine Similarity Mixed Model (Interaction)

```
## === Cosine Similarity Interaction Model (Same vs Previous Turn) ===

##                                 Estimate   Std. Error        df     t value
## (Intercept)                 0.6797851061 0.0064345344  683.1165 105.6463558
## turn                        0.0019961743 0.0008644214 1293.0000   2.3092605
## lag_typePrevious Turn      -0.0155788722 0.0090876890 1293.0000  -1.7142832
## turn:lag_typePrevious Turn  0.0004099441 0.0013471035 1293.0000   0.3043152
##                              Pr(>|t|)
## (Intercept)                0.00000000
## turn                       0.02108612
## lag_typePrevious Turn      0.08671638
## turn:lag_typePrevious Turn 0.76093673
```

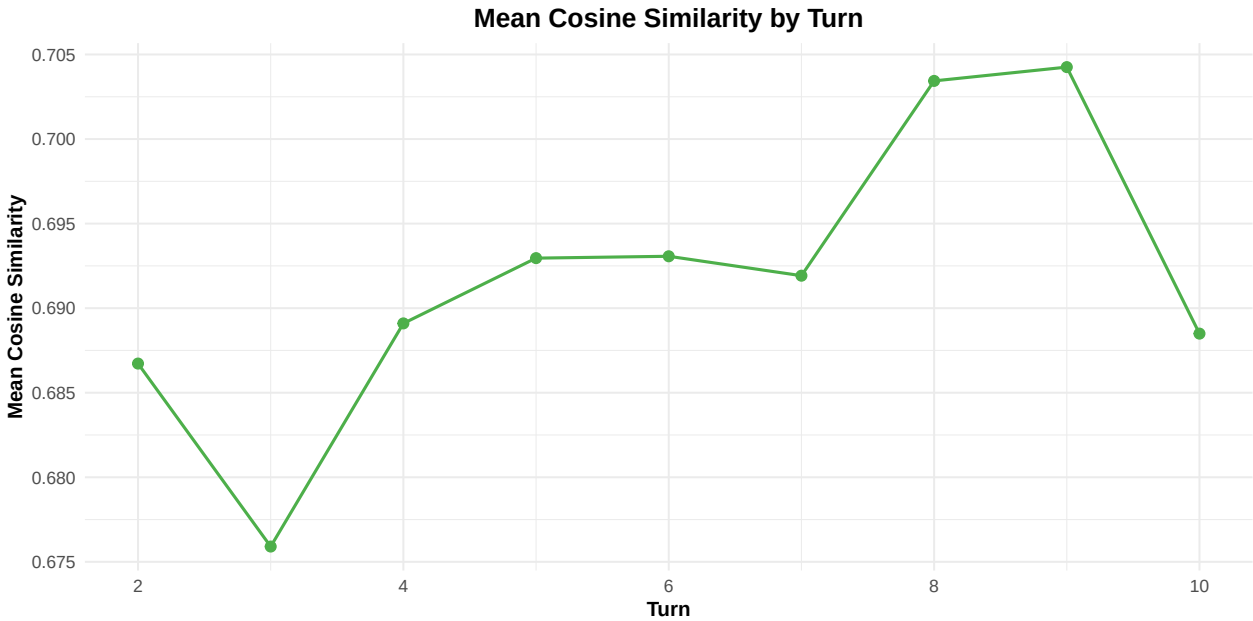
4.3 4.2b Comparison of Current vs Previous Turn Cosine Similarity

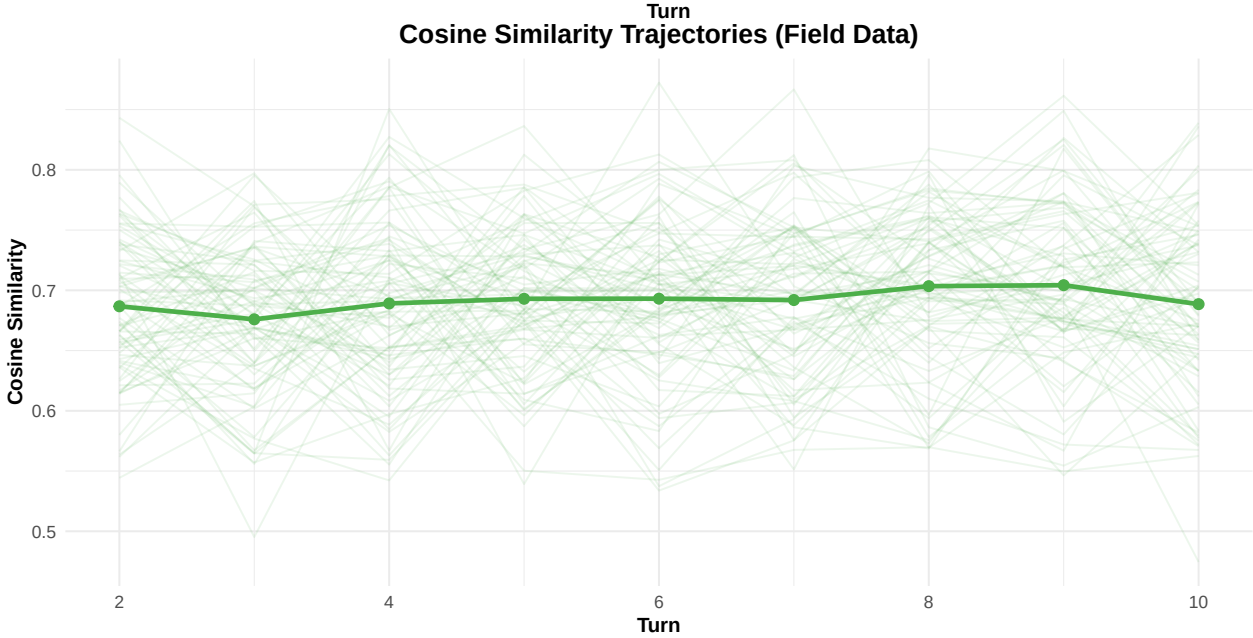


4.4 4.3 Cosine Similarity Linear Plot



4.5 4.4 Cosine Similarity Over Time





4.6 4.5 Cosine rubber band

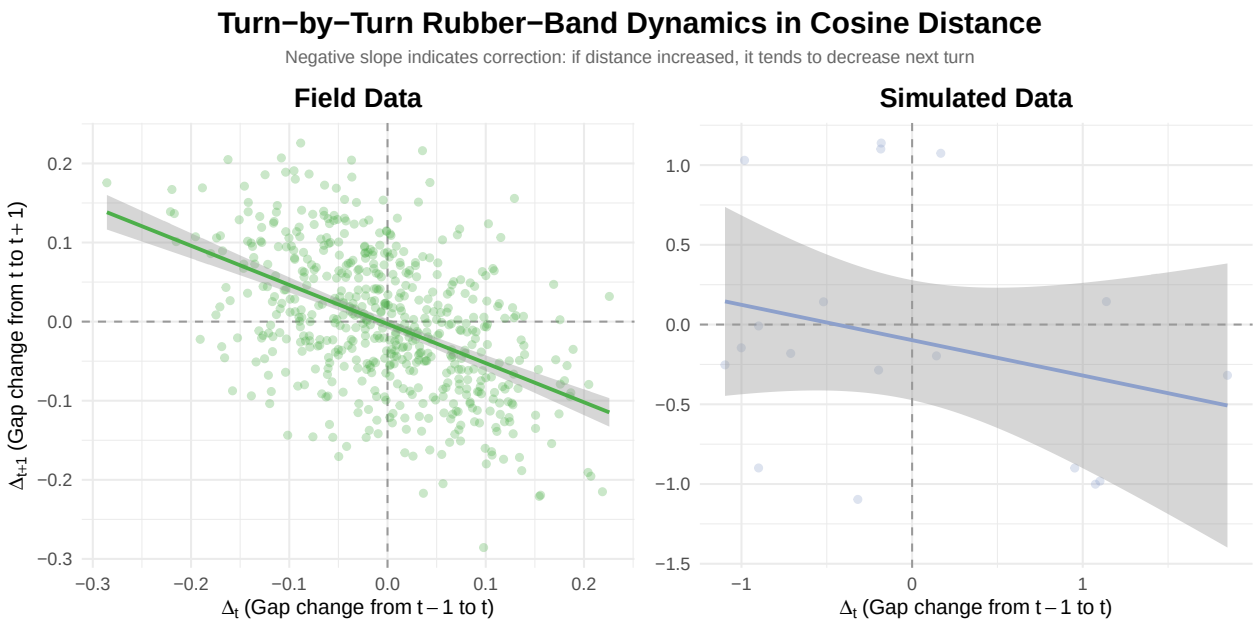
```
## === Turn-Level Rubber-Band Cosine Distance: Field Data ===

##                 Estimate  Std. Error     t value     Pr(>|t|)
## (Intercept) -0.003037546 0.003217033  -0.9442071 3.454674e-01
## dist_change -0.494722436 0.037627018 -13.1480640 1.223201e-34
```

4.7 4.6 Turn level Cosine Permutation test

```
## === Cosine Distance Turn-Level Permutation Test Results ===

## Observed slope: -0.4947

## Null mean: -3e-04
```

```
## Null SD: 0.0554
## Effect size (z): -8.93
## P-value: <2e-16
## # A tibble: 2 x 4
##   transition_type rb_slope correlation     n
##   <chr>              <dbl>       <dbl> <int>
## 1 Stage Boundary    -0.500      -0.507   162
## 2 Within-Stage      -0.497      -0.502   324
```

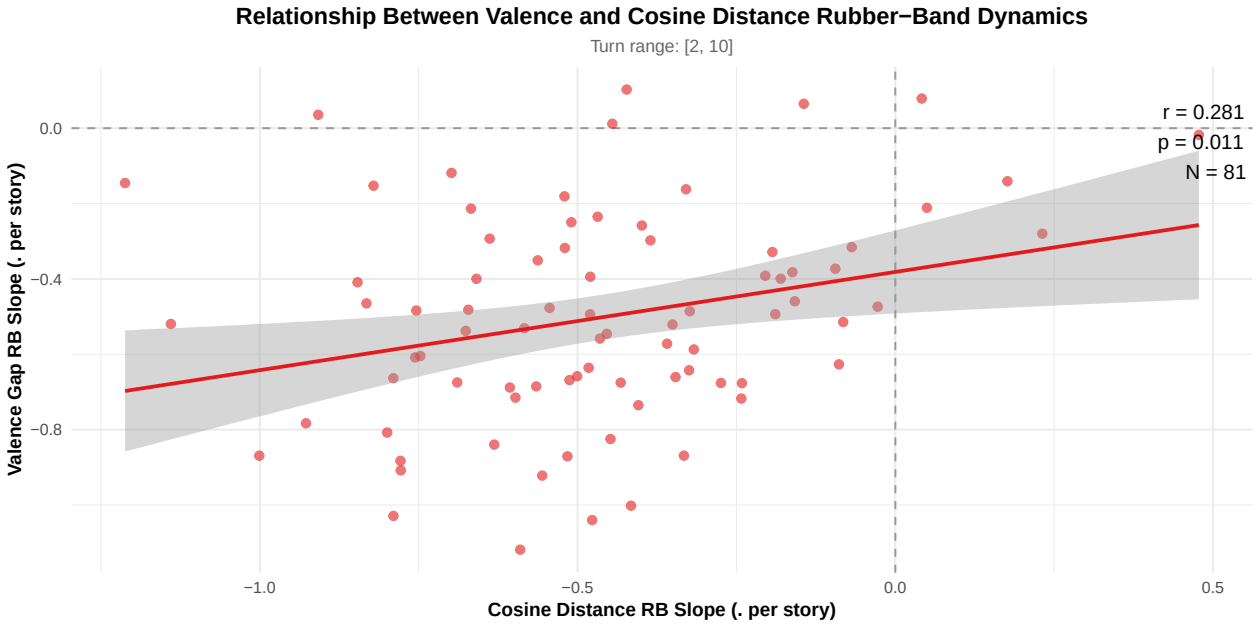
4.8 4.7 Turn Level Cosine RB Linear plot



4.9 4.8 Correlation Between Cosine Distance and Valence Rubber-Band

```
## === Correlation Between Valence and Cosine Distance Rubber-Band ===
## Turn range: [2, 10]
## N observations (turn-level): 567
## Turn-level correlation of gap changes (Δt): 0.032
##   Interpretation: Do valence and cosine gaps change in the same direction
##                   at the same turn across all stories?
## Turn-level correlation of next changes (Δt+1): 0.024
##   Interpretation: Do subsequent valence and cosine corrections correlate?
## N stories with >= 3 turns: 81
## N stories with >= 3 turns: 81
## STORY-LEVEL correlation of RB slopes: 0.281
##   Interpretation: Across stories, does valence RB strength predict cosine RB strength?
##                   Positive r means stories with strong valence correction
```

```
##                    also show strong cosine distance correction.
##   95% CI: [0.067, 0.47]
##   p-value: 0.0111
##   df: 79
## Shared variance (R²): 0.079
## Interpretation: 7.9 % of variation in valence RB slope
##                 is explained by cosine distance RB slope
```




```
##
## === Regression: Cosine RB Predicting Valence RB ===
##                   Estimate Std. Error   t value     Pr(>|t|)
## (Intercept)     -0.3815065  0.0553939 -6.887156 1.227194e-09
## cosine_rb_slope  0.2606403  0.1002001  2.601199 1.108681e-02
##
## === VALIDATION CHECK ===
## Correlation from cor(): 0.281
## Correlation from lm R²: 0.281
## These should match (they do if correlation is positive).
```

5 5. Supplementary: Interaction Effects

5.1 5.1 Starter Interaction Effects

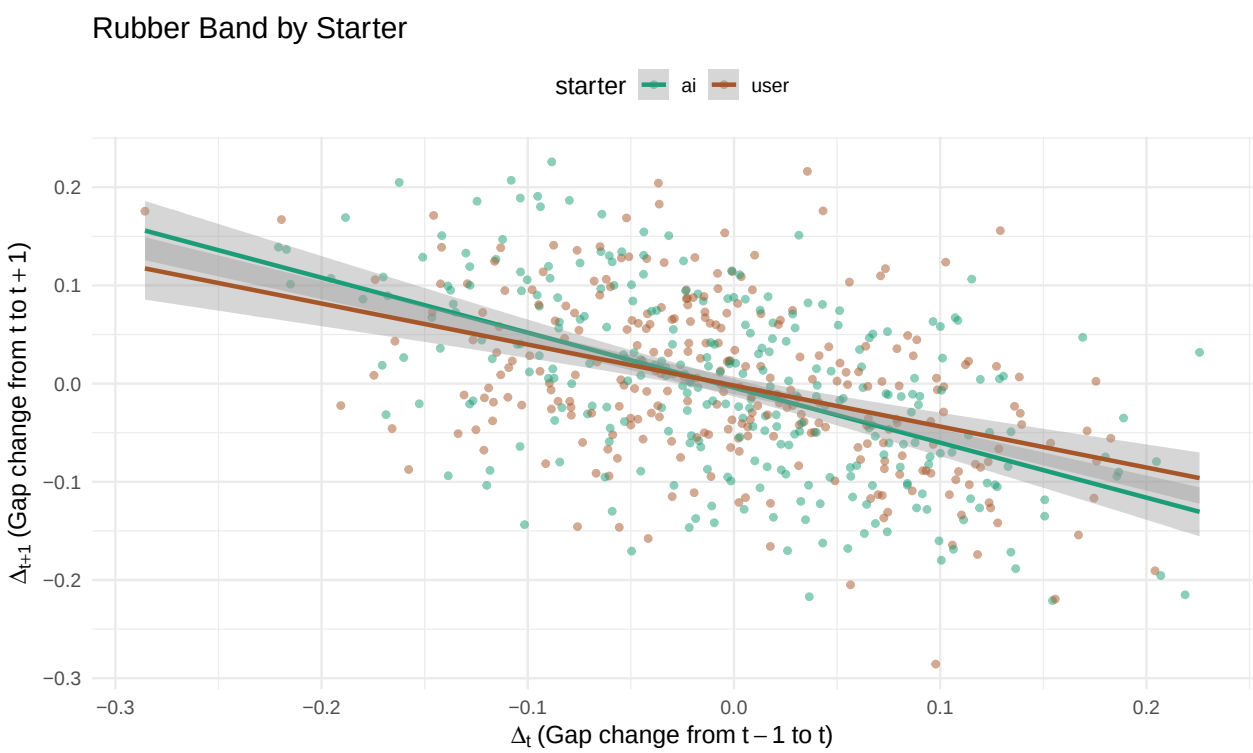
5.1.1 5.1.1 Valence Gap by Starter

```
## === Valence Rubber-Band: Starter Interaction ===
##                                     Estimate Std. Error     t value
## (Intercept)                      -0.013053027 0.02359380 -0.55323971
## gap_change                       -0.503752882 0.05067800 -9.94026807
```

```
## as.factor(starter)user            -0.001124884 0.03357459 -0.03350404
## gap_change:as.factor(starter)user -0.043265762 0.07036489 -0.61487713
##                                       Pr(>|t|)
## (Intercept)                       5.803189e-01
## gap_change                        1.469151e-21
## as.factor(starter)user            9.732845e-01
## gap_change:as.factor(starter)user 5.388843e-01
```

5.1.2 5.1.2 Cosine Distance by Starter

```
## === Cosine Distance Rubber-Band: Starter Interaction ===

##                              Estimate  Std. Error     t value     Pr(>|t|)
## (Intercept)             -0.004125568 0.004515614  -0.9136227 3.613062e-01
## dist_change             -0.559972708 0.051013408 -10.9769711 1.549421e-25
## starteruser              0.002173674 0.006425178   0.3383057 7.352590e-01
## dist_change:starteruser  0.142507964 0.075417588   1.8895853 5.932697e-02
```



5.1.3 5.1.3 Alignment (Cosine Similarity) by Starter

```
## === Cosine Similarity (Alignment): Starter × Turn Interaction ===

##                       Estimate  Std. Error    t value   Pr(>|t|)
## (Intercept)       0.6756066397 0.008710958 77.5582487 0.00000000
## turn              0.0023498453 0.001333588  1.7620479 0.07848258
## starteruser       0.0084613944 0.012395910  0.6825956 0.49508043
## turn:starteruser -0.0007161839 0.001897728 -0.3773901 0.70599420
```

5.2 5.2 LLM Type Interaction Effects

5.2.1 5.2.1 Valence Gap by LLM Type

```
## === Valence Rubber-Band: LLM Type Interaction ===

##                                 Estimate Std. Error    t value     Pr(>|t|)
## (Intercept)                   0.02551150 0.03562386  0.7161352 4.742069e-01
## gap_change                   -0.53724677 0.07244110 -7.4163250 4.493291e-13
## factor(llm_type)2            -0.03485347 0.05038391 -0.6917580 4.893765e-01
## factor(llm_type)3            -0.05069270 0.04712018 -1.0758171 2.824732e-01
## factor(llm_type)4            -0.06302335 0.04854157 -1.2983375 1.947068e-01
## gap_change:factor(llm_type)2  0.02948591 0.10705184  0.2754358 7.830832e-01
## gap_change:factor(llm_type)3  0.08204938 0.10159268  0.8076309 4.196466e-01
## gap_change:factor(llm_type)4 -0.04715818 0.09537181 -0.4944667 6.211709e-01
```

5.2.2 5.2.2 Cosine Distance by LLM Type

```
## === Cosine Distance Rubber-Band: LLM Type Interaction ===

##                                     Estimate  Std. Error  df     t value
## (Intercept)                   -0.0017963414 0.006847425 559 -0.26233824
## dist_change                   -0.5372565912 0.071885135 559 -7.47382048
## factor(llm_type)2             -0.0008625916 0.009682667 559 -0.08908616
## factor(llm_type)3             -0.0036395907 0.009065801 559 -0.40146376
## factor(llm_type)4             -0.0001667134 0.009331392 559 -0.01786586
## dist_change:factor(llm_type)2  0.1310224906 0.109626305 559  1.19517383
## dist_change:factor(llm_type)3  0.0099745635 0.098688567 559  0.10107111
## dist_change:factor(llm_type)4  0.0577425383 0.110369938 559  0.52317270
##                                    Pr(>|t|)
## (Intercept)                   7.931573e-01
## dist_change                   3.023375e-13
## factor(llm_type)2             9.290453e-01
## factor(llm_type)3             6.882321e-01
## factor(llm_type)4             9.857522e-01
## dist_change:factor(llm_type)2 2.325259e-01
## dist_change:factor(llm_type)3 9.195302e-01
## dist_change:factor(llm_type)4 6.010614e-01
```

5.2.3 5.2.3 Alignment (Cosine Similarity) by LLM Type

```
## === Cosine Similarity (Alignment): LLM Type × Turn Interaction ===

##                              Estimate  Std. Error       df     t value
## (Intercept)             0.6720050502 0.013399480 529.5861 50.15157707
## turn                    0.0020754132 0.001878479 644.0000  1.10483691
## factor(llm_type)2       0.0206740785 0.018949726 529.5861  1.09099616
## factor(llm_type)3       0.0118070092 0.017725846 529.5861  0.66609004
## factor(llm_type)4      -0.0012055763 0.018260413 529.5861 -0.06602131
## turn:factor(llm_type)2 -0.0007741944 0.002656571 644.0000 -0.29142622
## turn:factor(llm_type)3  0.0007692124 0.002484994 644.0000  0.30954291
## turn:factor(llm_type)4 -0.0005211406 0.002559936 644.0000 -0.20357567
##                              Pr(>|t|)
## (Intercept)            2.752150e-203
## turn                    2.696429e-01
## factor(llm_type)2       2.757707e-01
## factor(llm_type)3       5.056436e-01
```

```
## factor(llm_type)4       9.473858e-01
## turn:factor(llm_type)2  7.708192e-01
## turn:factor(llm_type)3  7.570087e-01
## turn:factor(llm_type)4  8.387494e-01
```